

RIO DE JANEIRO SANTOS DUMONT, Brazil

WMO: 837550

Lat: 22.910S

Lon: 43.163W

Elev: 3

StdP: 101.28

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	17.2	11.8	8.6	19.9	13.0	9.3	20.1	8.8	22.8	7.7	23.1	2.8	310	0.303

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.9	34.8	25.7	33.1	25.5	32.1	25.2	27.2	31.7	26.7	31.0	26.2	30.3	4.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.4	30.3	25.2	20.4	29.1	25.0	20.1	28.8	86.2	31.8	83.6	31.2	81.6	30.4	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.4	6.6	15.1	37.7	1.2	0.8	14.2	38.3	13.5	38.7	12.8	39.1	12.0	39.7		
			12.7	27.8	1.5	1.0	11.7	28.5	10.8	29.1	10.0	29.7	8.9	30.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.8	27.4	27.9	27.0	25.9	23.6	22.7	22.2	22.5	23.0	24.1	24.9	26.4	(6)
	DBStd	2.93	2.38	1.95	1.82	1.96	1.94	2.00	2.23	2.20	2.34	2.29	2.50	2.25	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	3	0	0	0	0	0	0	2	1	1	0	0	0	(9)
	CDD10.0	5391	540	500	528	476	420	380	377	387	390	437	447	509	(10)
	CDD18.3	2353	282	267	269	226	162	131	121	129	141	179	197	251	(11)
	CDH23.3	17077	2841	2803	2511	1736	710	444	402	494	626	998	1347	2167	(12)
(13)	CDH26.7	5094	1044	1057	776	428	113	66	66	95	139	229	382	699	(13)
(14)	Wind	WSAvg	3.1	3.2	3.2	3.0	2.8	2.7	2.8	3.0	3.3	3.4	3.3	3.3	(14)
(15) Precipitation	PrecAvg	1154	137	144	134	114	81	52	52	51	66	87	98	148	(15)
	PrecMax	1751	473	473	332	299	264	155	168	128	223	198	195	316	(16)
	PrecMin	583	31	0	29	18	9	2	2	1	1	3	33	60	(17)
	PrecStd	277	92	117	91	66	57	42	37	31	49	45	44	68	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	35.9	34.8	33.8	31.1	30.2	30.2	31.7	32.8	33.1	34.9	36.0	(19)
		MCWB	26.2	26.5	25.3	25.2	23.6	22.2	21.5	21.8	22.9	24.0	25.0	26.3	(20)
	2%	DB	34.1	33.9	32.2	31.2	29.1	28.2	28.2	29.1	29.9	30.8	32.1	33.1	(21)
		MCWB	26.4	25.9	25.8	25.1	23.3	21.9	21.4	21.5	22.4	24.0	24.6	25.5	(22)
	5%	DB	32.8	32.2	31.1	30.1	27.9	27.0	27.0	27.1	28.1	29.0	30.2	31.8	(23)
		MCWB	26.0	25.6	25.7	24.8	22.9	21.8	21.3	21.3	22.2	23.4	24.4	25.2	(24)
	10%	DB	31.1	31.2	30.1	29.0	26.8	25.9	25.2	25.9	26.8	27.8	28.9	30.1	(25)
		MCWB	25.5	25.6	25.2	24.2	22.6	21.6	20.9	21.0	21.8	23.1	23.8	24.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.9	28.0	27.7	26.7	25.6	24.4	24.1	23.5	24.7	26.0	26.4	27.2	(27)
		MCDB	33.4	33.4	31.4	30.5	28.2	27.2	26.4	28.0	29.0	29.8	31.0	32.4	(28)
	2%	WB	27.0	27.0	26.8	25.9	24.3	23.2	22.6	22.5	23.6	24.7	25.6	26.4	(29)
		MCDB	32.1	31.5	30.4	29.5	27.3	26.2	25.6	26.4	27.2	28.2	29.8	31.1	(30)
	5%	WB	26.4	26.5	26.3	25.3	23.6	22.5	21.9	22.0	22.8	24.1	24.9	25.7	(31)
		MCDB	31.0	30.8	29.7	28.6	26.5	25.5	25.3	25.6	26.2	27.5	28.8	30.1	(32)
	10%	WB	25.9	26.1	25.7	24.7	23.0	21.9	21.3	21.3	22.2	23.5	24.2	25.2	(33)
		MCDB	30.0	30.1	29.0	27.9	25.8	24.9	24.3	24.7	25.6	26.7	27.7	29.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.9	6.2	6.0	5.8	6.1	6.3	6.3	6.2	5.8	6.1	6.5	(35)
		MCDBR	8.9	8.5	7.7	7.6	7.3	8.0	8.6	8.6	9.1	8.3	8.9	9.2	(36)
	5% WB	MCWBR	3.6	3.4	3.3	3.3	3.3	3.4	3.6	3.5	3.7	3.6	3.6	3.6	(37)
		MCWBR	7.9	7.5	6.9	6.7	6.6	6.9	7.3	7.3	7.5	7.2	7.8	8.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.413	0.409	0.409	0.397	0.382	0.381	0.383	0.414	0.492	0.467	0.433	0.424	(40)
		taud	2.479	2.472	2.482	2.502	2.516	2.508	2.485	2.376	2.129	2.263	2.392	2.433	(41)
	Ebn at Noon		933	923	897	864	836	816	826	834	806	860	908	923	(42)
		Edh at Noon	118	117	111	101	92	89	93	112	155	142	128	124	(43)
(44) All-Sky Solar Radiation	RadAvg		6.05	6.28	5.36	4.70	3.72	3.48	3.60	4.27	4.60	5.17	5.34	5.90	(44)
	RadStd		0.88	0.64	0.44	0.33	0.20	0.25	0.31	0.32	0.58	0.60	0.44	0.53	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (4 neighbors)		N/A	N/A	-1.69	N/A	-1.40	-1.66	N/A	N/A	N/A	N/A

Nomenclature: See separate page